Introduction

Mixed-effects models (also called multilevel or hierarchical models) handle data with a clustered or nested structure: patients within hospitals, repeated measures within subjects, plants within plots. The model has **fixed effects** for population-level structure and **random effects** capturing cluster-specific deviations.

Prerequisites

Linear regression.

Theory

Simplest form: random intercept model

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim \mathcal{N}(0, \sigma_u^2),\ \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

u_i are subject-specific deviations; ε_{ij} are residuals. Intraclass correlation: $\rho = \sigma_u^2/(\sigma_u^2 + \sigma^2)$.

Extensions: random slopes (u_{1i}), nested or crossed groupings, non-normal responses (GLMM).

Assumptions

- Clusters are a random sample from a population of clusters.
- Random effects normal; residuals normal.
- Linearity, independence across clusters.

R Implementation

```r
library(lme4); library(lmerTest)

data(sleepstudy, package = "lme4")
head(sleepstudy)

# Random intercept
fit_ri <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
summary(fit_ri)

# Random intercept + slope
fit_rs <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
summary(fit_rs)

# Test random slope vs. intercept only
anova(fit_ri, fit_rs)
```

Output & Results

Fixed-effect table with estimates, SEs, df (approximated), p-values; random-effect variance / SD summary; likelihood-ratio test between nested random-effects structures.

Interpretation

"Each day of sleep deprivation increased reaction time by 10.5 ms (SE = 1.5, p < 0.001). Between-subject variation in intercept was substantial (SD = 37 ms), and random-slope comparison showed better fit for a model allowing slope variation (chi-squared = 42, p < 0.001)."

Practical Tips

- Specify random effects by the grouping factor.
- Use REML by default; switch to ML for likelihood-ratio tests between fixed-effect specifications.
- Singular fits often arise from over-parameterised random-effects structures; simplify.
- For p-values, use `lmerTest` (Kenward-Roger or Satterthwaite approximations) – standard `lme4` does not provide them.
- For hierarchical binary or count outcomes, use `glmer` or `glmmTMB`.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI